



Leveraging AI-enabled knowledge capabilities for new product performance: Evidence from manufacturing SMEs in emerging economies

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ABSTRACT

Emerging market manufacturing small and medium-sized enterprises (SMEs) struggle to translate artificial intelligence (AI) investments into innovation outcomes. This study investigates how AI-enabled knowledge capabilities drive new product performance by employing a sequential mixed-methods design with 261 manufacturing SMEs. Our findings reveal that AI-enabled knowledge capabilities operate through dual pathways: directly enhancing performance and indirectly through digital product innovation radicalness. Furthermore, this study identifies digital sustainability leadership as a critical boundary condition acting as both mediator and moderator, fundamentally reconfiguring how AI generates value. This study makes significant contributions to dynamic capabilities theory and the knowledge-based view. By positioning AI-enabled knowledge capabilities as critical dynamic capabilities, this research clarifies the transformation pathways through which AI generates competitive advantages and offer actionable insights for managers navigating resource-constrained environments.

1. Introduction

The contemporary manufacturing landscape confronts an existential imperative: traditional knowledge management no longer suffices to drive innovation in emerging markets. While the global knowledge economy generates over \$9 trillion annually (U.S. National Science Foundation, 2020), emerging market manufacturing SMEs struggle disproportionately to convert knowledge assets into new product performance (Kumar et al., 2020). A troubling implementation-performance gap persists: although AI fundamentally transforms knowledge processes, potentially unlocking \$2.3 trillion for emerging markets (World Economic Forum, 2023), most SMEs fail to realize the 40% higher success rates achieved by AI leaders (Rezaei, 2025). This disconnect reveals a critical theoretical void regarding how AI-enabled knowledge capabilities function as dynamic capabilities in resource-constrained contexts (Braganza et al., 2021; Priyono et al., 2020).

Prior research has examined knowledge management extensive, yet findings regarding the knowledge-performance relationship remain inconsistent. While some studies identify direct effects (Joshi & Sharma, 2004; MacCurtain et al., 2010) or mediating mechanisms like knowledge integration (De Luca & Atuahene-Gima, 2007; Su et al., 2013), others point to conflicting moderators such as technological turbulence

(Jin et al., 2019; Tsai & Hsu, 2014) or inverted U-shaped engagement effects (Yepeng et al., 2022). Furthermore, research on AI's role remains polarized. While AI expands knowledge depth (Khalid et al., 2025) and facilitates innovation stages (Cooper, 2024), it also introduces “dark side” risks, including knowledge hiding (Arias-Pérez & Huynh, 2023), leakage (Ritala et al., 2015), and algorithmic gatekeeping (Keegan et al., 2023; Papagiannidis et al., 2023). Existing frameworks, largely developed in pre-AI contexts (Grant, 1996; Teece, 2007), inadequately explain how AI transforms sensing and seizing mechanisms in volatile environments (Mikalef & Gupta, 2021; Raisch & Krakowski, 2021).

To address these gaps, this study grounds itself in dynamic capabilities theory and the knowledge-based view, integrating practice-theoretical perspectives (Whittington, 2006). This study proposes that AI-enabled knowledge capabilities do not automatically generate advantage but require specific transformation pathways (Schilke et al., 2018; Teece, 2018). Specifically, this study identifies digital product innovation radicalness - breakthroughs utilizing digital technologies (Nambisan et al., 2017; Pesch et al., 2021) - as a critical mediator. This responds to calls to explicate value creation mechanisms in AI-enabled systems (Kohtamäki et al., 2025; Sjödin et al., 2021). Furthermore, this research examines digital sustainability leadership as a boundary condition. As firms face dual pressures for digitalization and

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sustainability (Del Giudice et al., 2021; Fernando et al., 2019), this leadership construct aligns innovation with climate actions (Zeichiel et al., 2024; Zeng et al., 2024) and moderates performance outcomes (Sahoo et al., 2024), yet its intersection with dynamic capabilities remains underexplored (Parida et al., 2019; Warner & Wäger, 2019).

Consequently, this study investigates: (1) the role of AI-enabled knowledge capabilities in driving new product performance; (2) the mediating mechanism of digital product innovation radicalness; and (3) the moderating role of digital sustainability leadership. Analyzing data from 261 manufacturing SMEs, this study makes significant contributions to the literature. First, this study extends dynamic capabilities theory by demonstrating how AI transforms sensing, seizing, and reconfiguring processes in resource-constrained environments (Le et al., 2024; Najafi-Tavani et al., 2016; Najafi-Tavani et al., 2018), bridging the gap between computational and human-centric knowledge (Benbya et al., 2020; Cui & Wu, 2016; Zhou & Li, 2012). Second, this research clarifies the transformation pathways of AI capabilities, resolving inconsistencies in knowledge-performance relationships (Chang et al., 2014; Forés & Camisón, 2016). Third, this study establishes digital sustainability leadership as a pivotal boundary condition, advancing the integration of digital and sustainable transformation streams (Adomako et al., 2021; Jensen et al., 2007; Wei & Pardo, 2022).

2. Literature review

2.1. Emerging market manufacturing SMEs

Emerging market manufacturing SMEs operate within distinctive institutional and resource environments characterized by institutional voids, resource scarcity, market volatility, and regulatory uncertainties that fundamentally differentiate their innovation processes from developed market counterparts (Peng et al., 2008). Recent research highlights that emerging markets are more malleable than mature markets (Petry, 2025; Röell et al., 2024), evolving through the development of ecosystems of actors who exchange critical resources and knowledge. To successfully instigate market formation in such environments, firms must engage in disruptive market-shaping processes that extend beyond internal product development to encompass the “forging of ecosystem alliances” and the “catalyzing of institutional changes” (Sandvik et al., 2024). This dynamic perspective conceptualizes markets not as static contexts but as shapeable multi-actor value systems where purposive actions by firms can redesign exchange patterns, reconfigure networks, and reform institutions through collaboration with ecosystem stakeholders (Flaig & Ottosson, 2022; Nenonen et al., 2019). Specifically, Sandvik et al. (2024) argue that creating critical mass through ecosystem resources and orchestrating ecosystem dynamics are essential dimensions for overcoming barriers to commercializing disruptive innovations. These firms face acute competitive pressures stemming from limited access to advanced technologies, constrained financial resources, and underdeveloped knowledge infrastructures, yet simultaneously encounter unprecedented opportunities arising from rapidly growing domestic markets and increasing global integration (Kumar et al., 2020). Within this ecosystem perspective, emerging market manufacturing SMEs must perform distinct market-shaping roles, ranging from market drivers proposing new visions to market missionaries building collaborative networks, that require capabilities for discovering value potential and mobilizing resources across the broader market system (Flaig & Ottosson, 2022; Nenonen et al., 2019; Thomson et al., 2022). Theoretically, the dynamic capabilities perspective emphasizes that emerging market SMEs must develop superior sensing, seizing, and transforming capabilities to navigate environmental turbulence and resource constraints effectively (Teece, 2007), while the knowledge-based view posits that knowledge creation and integration capabilities become particularly critical when firms cannot rely on abundant financial or technological resources to drive innovation (Grant, 1996). AI-enabled knowledge capabilities

become especially salient in this ecosystem context, as they enhance firms' exploring and experimenting capabilities, enabling systematic probing of market receptiveness, recognition of system-wide resource linkages, and engagement of diverse stakeholders through data-driven insights (Nenonen et al., 2019). Empirical evidence demonstrates that emerging market manufacturing SMEs achieve innovation success through different pathways than developed market firms, with studies revealing that external knowledge absorption, entrepreneurial orientation, and adaptive capabilities play disproportionately important roles in resource-constrained contexts (Forés & Camisón, 2016; Le et al., 2024). However, these firms simultaneously face strategic dilemmas regarding knowledge sharing and protection, where collaborative innovation activities necessary for capability development create significant knowledge leakage risks that can undermine competitive advantages (Ritala et al., 2015). Understanding how emerging market manufacturing SMEs can leverage AI-enabled knowledge capabilities to overcome traditional resource constraints while managing knowledge governance challenges and effectively engaging with market ecosystems represents a critical theoretical and practical imperative, as these firms increasingly constitute the backbone of economic development yet struggle disproportionately to translate knowledge assets into new product performance outcomes that ensure survival and growth in intensifying global competition.

2.2. AI-enabled knowledge capabilities

Following Jensen et al.'s (2007) seminal distinction between Science-Technology-Innovation (STI) and Doing-Using-Interacting (DUI) modes of innovation, knowledge capabilities encompass both explicit, codified knowledge typical of STI modes (patents, formal R&D, scientific expertise) and tacit, experience-based knowledge characteristic of DUI modes (learning-by-doing, problem-solving, interactive learning). Building on foundational knowledge management research, scholars have distinguished between different types of knowledge accumulation capabilities, internal knowledge creation and external knowledge absorption, that differentially impact incremental versus radical innovation performance (Forés & Camisón, 2016; Su et al., 2013).

AI-enabled knowledge capabilities represent an organization's ability to integrate AI technologies with traditional knowledge management processes to generate, assimilate, and apply knowledge for enhanced operational and strategic outcomes. Building on Singh et al.'s (2025) conceptualization of knowledge capabilities as encompassing knowledge creation, infrastructure, transmission, and management culture, and Abou-Foul et al.'s (2023) AI capabilities framework, this study defines AI-enabled knowledge capabilities as an organization's ability to systematically collect, process, and apply AI-driven insights to enhance both explicit knowledge codification (STI mode) and tacit knowledge development (DUI mode) for decision-making processes, customer understanding, and innovation activities (Abou-Foul et al., 2023).

Critically, AI-enabled knowledge capabilities encompass three interconnected dimensions: (1) AI-driven data acquisition and processing capabilities that enable real-time capture and analysis of production, market, and customer information; (2) AI-facilitated knowledge integration capabilities that synthesize disparate knowledge sources across organizational boundaries; and (3) AI-augmented knowledge application capabilities that translate insights into actionable innovation activities (Mikalef & Gupta, 2021; Wamba et al., 2017). Specifically, AI technologies enable manufacturing SMEs to codify tacit production knowledge into explicit algorithms (converting DUI to STI), while simultaneously facilitating interactive learning through intelligent feedback systems (enhancing DUI modes) (Braganza et al., 2021; Mikalef & Gupta, 2021). Moreover, AI platforms enable diverse user interactions and role adoptions that shape knowledge creation processes (Wei & Pardo, 2022), while AI-enabled servitization facilitates execution-based services, expertise-driven roles, and hyper-personalization (Wahid et al., 2025).

The theoretical foundation for AI-enabled knowledge capabilities draws from the dynamic capabilities perspective (Teece, 2007) and knowledge-based view (Grant, 1996), which suggest that organizations must continuously reconfigure their knowledge assets to maintain competitive advantage. The integration of AI technologies with knowledge capabilities creates a dynamic capability that enables firms to sense market opportunities, seize knowledge-intensive innovations, and transform organizational learning processes (Mikalef & Gupta, 2021). This alignment supports Singh et al.'s (2025) assertion that knowledge capabilities serve as gateways to essential insights and best practices for informed decision-making.

Empirical evidence suggests that AI-enabled knowledge capabilities positively influence new product performance through enhanced knowledge processing and application capabilities. Recent research demonstrates that AI finds applications across idea generation, concept creation and testing, building robust business cases, and product design, engineering, development, and testing, with these applications showing significant benefits including more effective and productive voice-of-customer studies (Cooper, 2024). Research also demonstrates that AI-enabled data mining and analytics capabilities improve product innovation processes (Abou-Foul et al., 2023), while machine learning models optimize pricing and demand forecasting accuracy (Wirtz et al., 2019). Furthermore, AI-enabled knowledge capabilities facilitate digital product innovation radicalness by enabling firms to develop breakthrough innovations through advanced data analytics and predictive capabilities (Sjödin et al., 2021). The relationship between AI-enabled knowledge capabilities and digital sustainability leadership is supported by AI's capacity to optimize resource allocation and identify sustainable practices throughout the value chain (Vinueza et al., 2020), enabling firms to achieve both innovation and sustainability objectives simultaneously.

2.3. Digital product innovation radicalness

Following Garcia and Calantone's (2002) foundational typology of innovation radicalness as a continuum from incremental innovations (minor improvements to existing products) to radical innovations (fundamental changes requiring new knowledge and competencies), this study adopts Pesch et al.'s (2021) conceptualization of digital product innovation radicalness as the extent to which digital products represent breakthrough innovations that deliver completely new benefits to customers and make existing technologies obsolete. Importantly, digital products in manufacturing contexts refer to physical products enhanced with digital capabilities (smart sensors, connected machinery, intelligent components) rather than purely digital offerings, representing what Porter and Heppelmann (2014) term "smart, connected products" that bridge physical and digital realms. This conceptualization aligns with emerging research on AI-enabled product-service systems, where digital components enable continuous value creation through data-driven services, predictive maintenance, and performance optimization (Kohtamäki et al., 2025; Paschou et al., 2020). The theoretical framework differs from Christensen's (2015) disruptive innovation framework, which focuses on market displacement patterns; instead, this study examines technological discontinuity and customer value creation dimensions (Gatignon et al., 2002), specifically focusing on the degree to which new products incorporate digital technologies, including embedded sensors, IoT connectivity, data analytics capabilities, and smart functionalities, to create offerings that significantly depart from existing market solutions in terms of technological newness and customer benefit novelty (Nambisan et al., 2017).

2.4. Digital sustainability leadership

Building on Parida et al.'s (2019) conceptualization of digital sustainability as the integration of digital technologies to achieve environmental, social, and economic sustainability outcomes, digital

sustainability leadership represents the strategic capability to guide organizations in leveraging digital innovations for sustainable value creation (Wang et al., 2025). Specifically, digital sustainability leadership encompasses an organization's strategic orientation toward integrating digital technologies with sustainability goals to create long-term value for stakeholders while addressing environmental and social challenges. Unlike generic leadership forms, digital sustainability leadership uniquely combines digital competence with sustainability vision, enabling leaders to navigate the complex trilemma of technological advancement, environmental stewardship, and economic viability (Bouncken et al., 2021). In the context of emerging market manufacturing SMEs, digital sustainability leadership becomes particularly critical as these firms face intensifying pressures from global supply chains, regulatory frameworks, and conscious consumers demanding sustainable practices (Fernando et al., 2019), while simultaneously pursuing digital transformation initiatives. This dual imperative creates opportunities for synergistic value creation, where digital technologies enable resource optimization, waste reduction, and circular economy practices, while sustainability commitments drive innovation in digital product design and business model transformation (Del Giudice et al., 2021). Digital sustainability leadership serves both as a direct driver of new product performance, by aligning innovations with market demands for sustainable solutions, and as a boundary condition that shapes how digital product innovation radicalness translates into performance outcomes, suggesting potential moderated mediation effects in the AI-knowledge-performance relationship.

2.5. New product performance

New product performance constitutes a multidimensional construct encompassing the achievement of financial, market, and strategic objectives through successful product launches, including revenue growth, market share expansion, profitability improvements, and customer satisfaction outcomes (Story et al., 2015). The theoretical significance of new product performance extends beyond immediate financial returns to represent a firm's capacity to leverage innovation capabilities for sustained competitive advantage, particularly in resource-constrained emerging market contexts where innovation success becomes a critical determinant of organizational survival and growth (Kumar et al., 2020). Importantly, emerging markets are increasingly recognized as malleable, dynamic value systems that evolve through ecosystems of actors who exchange critical resources and knowledge, rather than static competitive arenas (Nenonen et al., 2019). In these contexts, new product performance reflects not only individual firm outcomes but also the firm's ability to engage in market-shaping activities - purposive actions to re-design exchange relationships, re-configure network structures, and enhance value creation and realization for multiple stakeholders within the market system (Flaig & Ottosson, 2022; Nenonen et al., 2019). Manufacturing SMEs in emerging markets must navigate diverse actor roles within their ecosystems, from market drivers who propose innovation visions and commit resources, to market catalyzers who set standards, to market missionaries who foster collaborative networks, each contributing distinctly to new product commercialization success (Flaig & Ottosson, 2022).

Empirically, research demonstrates that new product performance outcomes vary substantially based on knowledge integration mechanisms (De Luca & Atuahene-Gima, 2007), innovation radicalness (Chang et al., 2014), and organizational capabilities that enable effective translation of knowledge assets into market-relevant innovations (Najafi-Tavani et al., 2018). From an ecosystem perspective, successful commercialization requires alignment between technological capabilities, ecosystem configuration maturity, and business model design that incentivizes desired collaborative behaviors across network partners (Thomson et al., 2022). Firms must develop capabilities to manage not only their own resources but also system-wide resources, making access to distributed knowledge and complementary assets as critical as

resource ownership (Nenonen et al., 2019). However, the literature reveals inconsistent findings regarding the antecedents and mediating pathways through which firms achieve superior new product performance, with some studies identifying direct effects of knowledge capabilities (MacCurtain et al., 2010) while others emphasize the necessity of mediating mechanisms such as absorptive capacity and market orientation (Najafi-Tavani et al., 2016). In manufacturing SME contexts, where competitive pressures intensify and resource constraints limit innovation experimentation (Adomako et al., 2021), understanding the precise mechanisms through which AI-enabled capabilities and digital innovation strategies translate into superior new product performance outcomes represents a critical theoretical and practical imperative for advancing innovation management scholarship. This understanding must account for how AI-enabled knowledge capabilities facilitate market-shaping by enabling firms to discover value potential through exploring system-wide resources and experimenting with market propositions, while mobilizing resources through expressing compelling visions and engaging ecosystem actors in collective value creation (Nenonen et al., 2019; Thomson et al., 2022).

3. Conceptual background and hypothesis development

3.1. Study 1: thematic analysis qualitative study to develop the conceptual model

This research employs a sequential mixed-methods design combining qualitative exploration with quantitative validation to comprehensively examine AI-enabled knowledge capabilities in emerging market manufacturing SMEs (Creswell & Clark, 2017; Venkatesh et al., 2016). Given the limited understanding of how AI-enabled knowledge capabilities manifest and operate in resource-constrained emerging market contexts, Study 1 employs qualitative thematic analysis to explore how these capabilities emerge in practice, their connections to digital innovation processes, and their influence on performance outcomes (Edmondson & Mcmanus, 2007; Gioia et al., 2013). Through systematic analysis of managerial experiences, this exploratory phase develops propositions that form the foundation for our conceptual model. Subsequently, Study 2 employs large-scale survey research with complementary analytical techniques to empirically validate these propositions. This sequential integration of qualitative depth and quantitative breadth ensures both contextual richness and statistical generalizability (Venkatesh et al., 2016).

3.1.1. Research method

We conducted comprehensive semi-structured interviews with sixteen manufacturing SME managers to investigate the impacts of potential factors when investigating the correlation between AI-enabled knowledge capabilities and new product performance in emerging market manufacturing SMEs. The qualitative study assists us in formulating the conceptual model in the first stage. While interview methodologies present certain constraints including potential interviewer bias, they enable focused exploration of complex organizational phenomena and capture participants' direct experiences with emerging AI technologies in manufacturing contexts (Gioia et al., 2013). Our sequential mixed-methods design adheres to established principles for methodological alignment in organizational studies (Edmondson & Mcmanus, 2007), where theoretical domain maturity informs methodological selection. Given that research on AI-enabled knowledge capabilities in emerging market SMEs represents an intermediate theoretical domain with evolving explanations and developing frameworks, a mixed-methods approach offers suitable methodological alignment (Fisher & Aguinis, 2017).

Our purposive sampling approach ensured participants could contribute meaningful insights addressing our research questions regarding AI integration in manufacturing SME operations. We implemented a criterion-based sampling strategy with clearly articulated

inclusion parameters to secure a representative sample (Ngo et al., 2020). Participants were chosen according to four specific criteria: (1) employment in manufacturing SMEs for minimum 10 years, ensuring adequate industry familiarity; (2) direct engagement in AI implementation initiatives or digital innovation projects; (3) representation across various organizational levels and manufacturing sectors to capture diverse viewpoints; and (4) experience with technology-enabled business practices as confirmed through initial screening interviews. We employed stratified purposive sampling to ensure balanced representation across gender, experience levels, and functional roles within manufacturing and technology domains (Morse, 2015).

The interviewed participants included nine males and seven females, with mean age of 41.3 years. Their average manufacturing experience was 13.7 years, with 87% having worked in SME environments for over 12 years. Among these participants, six held managerial positions, seven were in operational roles, and three occupied specialized technical functions related to AI or digital innovation. All participants reported involvement in technology adoption initiatives, with varying AI exposure ranging from basic implementations to advanced deployment projects. Interviewees represented diverse functional areas including operations management, product development, technology implementation, and strategic planning, capturing varied perspectives on how AI-enabled knowledge capabilities manifest across different SME contexts. Manufacturing sectors represented included electronics assembly (Interviewees 1, 9), textile production (Interviewees 2, 10), automotive parts (Interviewees 3, 11), food processing (Interviewees 4, 12), chemical manufacturing (Interviewees 5, 13), furniture production (Interviewees 6, 14), machinery manufacturing (Interviewees 7, 15), and plastic products (Interviewee 8, 16).

The semi-structured interview protocol contained six open-ended questions exploring experiences with AI-enabled knowledge capabilities, digital innovation processes, sustainability leadership practices, and perceived performance impacts. Our interview framework was systematically developed following established qualitative research standards (Creswell & Poth, 2016). Each 35–45 min interview was conducted individually via video conference, audio-recorded with consent, and transcribed verbatim. Participants received a research summary report as acknowledgment for their participation.

3.1.2. Data analysis

We analyzed data employing thematic analysis methodologies (Braun & Clarke, 2006), advancing from initial coding through pattern coding to theoretical coding, and utilized member validation and analyst triangulation to strengthen credibility. Our analytical process followed systematic qualitative coding procedures (Gioia et al., 2013). Initially, we developed a comprehensive coding framework with operational definitions of key constructs drawn from AI capabilities and manufacturing innovation literature. During initial coding, we identified 156 first-order codes across all transcripts using NVivo 15, with each transcript independently analyzed by two researchers ($\kappa = 0.91$, demonstrating strong inter-coder agreement). Through pattern coding, these first-order codes were integrated into 21 s-order themes via iterative researcher discussions and theoretical comparison. During theoretical coding, these themes were synthesized into four core theoretical dimensions: AI-enabled knowledge capabilities, digital product innovation radicalness, digital sustainability leadership, and new product performance.

3.1.3. Qualitative data analysis

Following established qualitative research protocols, our interview data underwent a systematic three-phase analytical process (Strauss & Corbin, 1998). Analysis of interview transcripts revealed four primary thematic clusters: AI-enabled knowledge capabilities (47 references across 16 interviews), digital product innovation radicalness (41 references across 15 interviews), digital sustainability leadership (39 references across 14 interviews), and new product performance (52

references across 16 interviews).

Concerning AI-enabled knowledge capabilities, participants described diverse levels of current adoption and future aspirations. The limited AI adoption theme encompassed basic digital tools usage (21 references) and cost-constraint implementation challenges. Emerging AI integration featured pilot projects and gradual capability building strategies. Knowledge enhancement benefits highlighted improved market insights and competitive intelligence gains through technology utilization.

The digital product innovation radicalness theme was characterized by participants' emphasis on incremental digital improvements (19 references across 12 interviews) and breakthrough innovation attempts (16 references across 10 interviews). Innovation outcomes emerged as critical factors, with participants noting enhanced product features and market differentiation capabilities.

Digital sustainability leadership was distinguished by compliance-driven practices (17 references across 9 interviews) and proactive sustainability innovation (13 references across 8 interviews). Leadership positioning included strategic initiatives and digital-enabled green solutions implementation.

New product performance outcomes encompassed market acceptance challenges (18 references across 11 interviews), export market success (15 references across 9 interviews), and innovation-performance linkage (19 references across 10 interviews).

Our analytical framework integrated inductive and abductive reasoning approaches (Timmermans & Tavory, 2012). While we initiated with inductive coding to enable themes to emerge naturally from the data, we subsequently employed abductive inference to connect these emergent patterns to existing theoretical frameworks in AI capabilities and manufacturing innovation literature. This methodology enabled us to remain receptive to novel insights while positioning our discoveries within established literature on AI-enabled organizational capabilities and digital innovation practices.

These findings (Table 2) provided substantial evidence supporting the interconnections between AI knowledge capabilities, digital innovation, and performance outcomes while revealing complex relationships between technological capabilities, sustainability leadership, and market performance in emerging market manufacturing SME contexts across diverse industrial sectors, informing our conceptual model for quantitative validation. The diversity in responses across different manufacturing sectors demonstrated the broad applicability of our theoretical framework while highlighting sector-specific implementation considerations, particularly in the AI-enabled knowledge capabilities domain, emphasizing both current constraints and future opportunities for emerging market SMEs.

3.2. Foundational theories and research framework

The dynamic capabilities perspective posits that firms achieve competitive advantage through their ability to sense, seize, and transform opportunities by reconfiguring internal and external competencies in response to rapidly changing environments (Teece, 2007). This theoretical framework has been particularly relevant in understanding how organizations develop and deploy capabilities that enable continuous adaptation and innovation (Eisenhardt & Martin, 2000). Within the context of emerging market manufacturing, the dynamic capabilities theory provides a lens to explore how firms can leverage dynamic capabilities to navigate uncertainty and capitalize on technological opportunities (Peng et al., 2008).

The knowledge-based view complements the dynamic capabilities theory by emphasizing knowledge as the most strategically significant resource for firm performance and competitive advantage (Grant, 1996). The knowledge-based view suggests that firms exist primarily to create, transfer, and apply knowledge, with superior knowledge integration capabilities serving as the foundation for sustained competitive advantage (Kogut & Zander, 1992). In manufacturing contexts, the

knowledge-based view explains how firms can transform knowledge assets into innovative products and processes that drive performance outcomes (Chen & Huang, 2009).

The integration of AI technologies represents both a dynamic capability and a knowledge-intensive resource that enables firms to enhance their sensing, learning, and reconfiguration abilities (Mikalef & Gupta, 2021). AI-enabled knowledge capabilities allow organizations to process complex information, identify patterns, and generate insights that support strategic decision-making and innovation processes (Davenport & Ronanki, 2018). These capabilities reflect the foundational logic of dynamic capabilities by enabling firms to continuously sense market opportunities and reconfigure knowledge assets in response to changing competitive conditions.

From a knowledge-based perspective, AI-enabled knowledge capabilities enhance firms' ability to create, acquire, and apply knowledge more effectively, thereby improving new product development outcomes (Liu et al., 2025; Wamba et al., 2017). The integration of AI with knowledge processes enables firms to accelerate learning cycles, improve decision quality, and develop more innovative products that meet evolving market demands. However, consistent with both theoretical perspectives, this study argues that AI-enabled knowledge capabilities must be complemented by appropriate organizational mechanisms and leadership capabilities to realize their full potential.

Building on these theoretical foundations and as illustrated in the conceptual model (Fig. 1), this research positions AI-enabled knowledge capabilities as a dynamic capability that enhances new product performance through digital product innovation radicalness. We propose that digital product innovation radicalness serves as a mediating mechanism that translates AI-enabled knowledge capabilities into superior performance outcomes. Furthermore, consistent with the contingency perspective embedded within dynamic capabilities theory, we posit that digital sustainability leadership plays a moderating role, enhancing the effectiveness of digital innovation in driving new product performance outcomes.

3.3. Hypothesis development

3.3.1. AI-enabled knowledge capabilities and new product performance

The relationship between knowledge capabilities and new product performance is well-established in the literature (Table 1), with empirical evidence demonstrating positive effects across various manufacturing contexts. Joshi and Sharma (2004) found that customer knowledge development capabilities enhance new product performance through improved market understanding, while De Luca and Atuahene-Gima (2007) showed that knowledge integration mechanisms mediate the relationship between market knowledge and product innovation performance. Chang et al. (2014) further demonstrated that knowledge creation capabilities enable firms to develop both novel and appropriate products.

However, critical analysis reveals significant limitations in existing research. Studies show inconsistent findings regarding knowledge effects, with Tsai and Hsu (2014) finding that competitive intensity weakens knowledge integration effectiveness, and Jin et al. (2019)

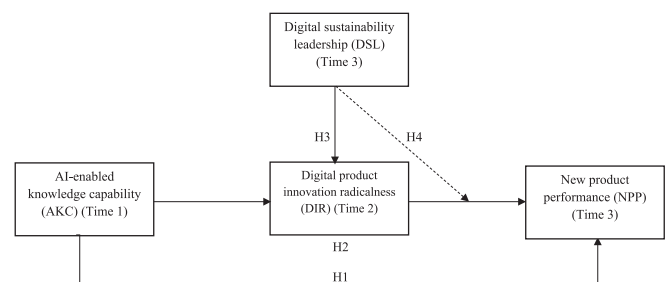


Fig. 1. Conceptual model.

Table 1

Summary of relevant studies to the literature of knowledge-related capabilities and product performance.

Authors	Independent variables	Dependent variables	Mediators	Moderators	Setting and Methodology	Foundational theories	Main findings relevant to knowledge-related capabilities and product performance
This study	AI-enabled knowledge capability	New product performance	Digital product innovation radicalness	Digital sustainability leadership	Emerging market manufacturing SMEs; PLS-SEM, fsQCA	Knowledge-based view, Dynamic capabilities theory	(1) AI-enabled knowledge capabilities directly enhance new product performance; (2) Digital product innovation radicalness mediates the relationship between AI-enabled knowledge capabilities and new product performance; (3) Digital sustainability leadership moderates the digital product innovation radicalness-new product performance relationship; (4) Digital product innovation radicalness mediates the relationship between digital sustainability leadership and new product performance.
(Joshi & Sharma, 2004)	Provision of resource slack; Intelligent-failure reward system; Creation of crossfunctional new product development teams; Integration mode of conflict resolution; Championing the organizational goal of product leadership; Project members' goal of product leadership	New product performance	Customer knowledge development	–	169 marketing managers of manufacturing firms in Canada	–	(1) Customer knowledge development enhances new product performance.
(De Luca & Atuahene-Gima, 2007)	Cross-functional collaboration; market knowledge dimensions	Product innovation performance	Knowledge integration mechanisms	Knowledge integration mechanisms	363 high-tech Chinese firms; EFA, CFA, and regression	Contingency theory; Knowledge-based view	(1) Market knowledge specificity and cross-functional collaboration affect product innovation performance through knowledge integration mechanisms. (2) The effect of market knowledge depth is partially mediated, market knowledge breadth has a direct, unmediated effect on product innovation performance. (3) The effects of market knowledge depth and cross-functional collaboration on product innovation are negatively moderated by knowledge integration mechanisms.
(MacCurtain et al., 2010)	Educational diversity; tenure diversity; age diversity; functional diversity; top management team trustworthiness	New product performance	Relexivity; knowledge sharing	–	39 indigenous software firms in Ireland		(1) Knowledge sharing and task reflexivity had direct effects on market new product performance

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Table 1 (continued)

Authors	Independent variables	Dependent variables	Mediators	Moderators	Setting and Methodology	Foundational theories	Main findings relevant to knowledge-related capabilities and product performance
(Yao et al., 2013)	Knowledge Complementarity	New product performance	Knowledge Absorption Effectiveness	Organizational Departmentalization; Organizational Culture: Learning Culture	119 international joint ventures in China; SEM	–	(1) Knowledge complementarity influences new product performance through the full mediation of knowledge absorption effectiveness. (1) competitive intensity weakens the effect of cross-functional collaboration on new product performance. (2) knowledge integration mechanisms mediate the negative effect of competitive intensity on the cross-functional collaboration–new product performance relationship.
(Tsai & Hsu, 2014)	Cross-functional collaboration	New product performance	Knowledge integration mechanisms	Competitive intensity	182 manufacturing firms in Taiwan; regression	structural contingency theory, fit-as-moderation and fit-as-mediation	(1) Knowledge creation helps research and development personnel create novel and appropriate products, which in turn increase new product success. (2) New product performance is affected by both novelty and appropriateness (1) The three forms of customer involvement are driven by different factors. (2) Furthermore, the impact of customer involvement on product performance is contingent upon the firm's technological capability, and the contingent effect also varies across different forms of customer involvement.
(J. Chang et al., 2014)	Knowledge creation	New product performance	Novelty; appropriateness	–	215 Taiwanese manufacturing firms; regression	–	(1) a positive relationship among market orientation, marketing capability, and new product performance. (2) Absorptive capacity positively moderates the relationship between market orientation and firms' newproduct performance (1) Product innovation capabilities are important predictors of new product performance. (2) Process innovation capabilities are important predictors of new product performance.
(Cui & Wu, 2016)	Nature of customer knowledge (customer need heterogeneity; customer need tacitness); knowledge management strategy (market exploitation, market exploration); knowledge management implementation (interfunctional coordination, strategic flexibility)	New product performance	Customer involvement (customer involvement as an information source, customer involvement as co-developers, customer involvement as innovators)	Technological capability	245 firms in North America; regression	–	(1) Market newness has a stronger positive effect on product performance (continued on next page)
(Najafi-Tavani et al., 2016)	Market orientation (market intelligence generation, market intelligence dissemination, responsiveness to market intelligence); market orientation x marketing capability; marketing capability	New product performance	Absorptive capacity	–	188 manufacturing firms in Sweden	Dynamic capabilities theory	
(Najafi-Tavani et al., 2018)	Absorptive capacity, collaborative innovation networks	New product performance	Product innovation capability; process innovation capability	–	258 Iranian high and medium technology manufacturin firms; SEM	–	
(Jin et al., 2019)	Technological newness; market newness	New product performance	–	Market knowledge breath; market knowledge tacitness	New high-tech ventures in China	–	

Table 1 (continued)

Authors	Independent variables	Dependent variables	Mediators	Moderators	Setting and Methodology	Foundational theories	Main findings relevant to knowledge-related capabilities and product performance
(Adomako et al., 2021)	R&D intensity	New product performance	International R&D teams	Knowledge creation process	201 Ghanaian firms	Knowledge-based view	than technological newness. (2) Market knowledge breadth enhances the effect of technological newness on product performance, whereas market knowledge tacitness weakens the effect of technological newness but enhances the effect of market newness on new product performance. (1) the use of international R&D teams mediates the relationship between R&D intensity and new product performance. (2) the use of international R&D teams improves new product performance and that this linkage is amplified when the knowledge creation process inside the firm is stronger. (1) there is an inverted U-shaped relationship between supplier engagement and new product innovation performance. (2) Supplier engagement positively impacts new product innovation performance, and knowledge sharing has an inverted U-shaped relationship with new product innovation performance. (3) when firms have a high-level rather than a low-level knowledge orchestration capability, knowledge sharing has a significant mediating role. (4) knowledge orchestration capability positively moderates the inverted U-shaped relationship between knowledge sharing and new product innovation performance, and the inflection point of the inverted U-shaped curve is shifted upward when knowledge orchestration capability is high. (1) user participation exerts an inverted U-shaped relationship with knowledge creation, which positively affects new (continued on next page)
(Yepeng et al., 2022)	Supplier engagement using social media	New product innovation performance	Knowledge sharing	Knowledge orchestration capability	151 Chinese firms; SEM	–	
(Wu et al., 2023)	Social media-based user participation	New product performance	Knowledge creation	Network capability	Chinese firms; SEM	–	

Table 1 (continued)

Authors	Independent variables	Dependent variables	Mediators	Moderators	Setting and Methodology	Foundational theories	Main findings relevant to knowledge-related capabilities and product performance
(Ozdemir et al., 2024)	Deployment of customer analytics	New product performance	Internal capabilities; knowledge integration mechanisms	Market knowledge breadth; exploitation learning, technological turbulence	249 high and medium tech firms based in China	Knowledge-based view; contingency theory	product performance. (2) when firms have high-level rather than low-level network capability, knowledge creation plays a significant mediating role (1) while some internal contingencies (such as exploitative learning strategy and market knowledge breadth) negatively moderate the effect of customer analytics on new product performance, others (such as internal capability and knowledge integration mechanisms) mediate its effect on performance (1) both experiential and vicarious learning from failure enhance new product performance. (2) as collective efficacy increases, the positive effect of experiential learning from failure on speed to market is strengthened. However, the positive effect of vicarious learning from failure on product innovativeness is weakened. (3) NPD teams can benefit from experiential and vicarious learning from failure to improve new product performance but must pay attention to the double-edged effect of collective efficacy
(Tao et al., 2025)	Experiential learning from failure; vicarious learning from failure	New product performance (speed to market, product innovativeness)	–	Collective efficacy	152 new product development teams in Chinese high-tech SMEs	Social cognitive theory	

revealing that knowledge tactness can actually harm performance outcomes. Additionally, [Yepeng et al. \(2022\)](#) identified an inverted U-shaped relationship between knowledge sharing and performance, suggesting potential negative effects of excessive knowledge activities. Moreover, recent research further reveals AI's dark side, including negative or unintended consequences grouped into work nature issues, conflicts and effects, and responsibility concerns ([Papagiannidis et al., 2023](#)), alongside knowledge hiding dynamics where AI capability negatively affects outbound open innovation when partner trustworthiness is low ([Arias-Pérez & Huynh, 2023](#)), and network-yielding dark force consequences such as algorithmic gatekeeping and dehumanization ([Keegan et al., 2023](#)).

The integration of AI technologies addresses these limitations by fundamentally transforming knowledge processing capabilities. Unlike traditional knowledge management that relies on human cognitive limitations, AI-enabled knowledge capabilities process vast datasets simultaneously, identify complex patterns, and enable real-time learning ([Abou-Foul et al., 2023](#)). AI applications across new product development, from idea generation and concept testing to optimal

product design, achieve remarkable reductions in development and testing times, more appealing product ideas and concepts, and more effective voice-of-customer studies ([Cooper, 2024](#)). The mechanism operates through enhanced sensing capabilities for opportunity identification, improved knowledge integration across organizational silos, and accelerated learning cycles for rapid product iteration ([Davenport & Ronanki, 2018](#)).

Despite these theoretical advantages, empirical evidence specifically examining AI-enabled knowledge capabilities remains limited. [Ozdemir et al. \(2024\)](#) found that customer analytics deployment positively impacts performance through knowledge integration, though organizational contingencies may constrain effectiveness. However, [Cooper's \(2025\)](#) study reveals that SMEs have not implemented AI across most of the 13 possible application areas in new product development to a great extent, with modest performance results averaging 27% improvement, and readiness-to-adopt AI remains not strong due to high acquisition costs, business case challenges, cybersecurity and IP risks, and recent AI failures. The critical gap lies in understanding how AI fundamentally transforms rather than merely augments traditional knowledge

Table 2

Thematic analysis and construct development from interview data.

Aggregate theoretical dimensions	Second-order themes	First-order codes	Representative quotes
AI-enabled knowledge capabilities	Limited AI adoption	- Basic digital tools usage - Cost-constraint implementation - Manual knowledge processes - Technical skill gaps	“We use Excel and basic CRM, no budget for AI yet” (Interviewee 2) “Still rely on personal networks for market intelligence” (Interviewee 8) “AI sounds expensive for small companies like us” (Interviewee 14)
	Emerging AI integration	- Pilot AI projects - Gradual capability building - Learning-by-doing approach - External support utilization - Cross-functional collaboration	“Started with chatbot, now testing demand forecasting” (Interviewee 1) “Learning AI tools through online courses and trials” (Interviewee 5) “Government program helped us pilot AI for production planning” (Interviewee 11)
	Knowledge enhancement benefits	- Improved market insights - Better customer understanding - Competitive intelligence gains - Predictive analytics capabilities - Data-driven decision making	“AI gives us 70% accuracy in seasonal demand prediction” (Interviewee 3) “Better understanding of export customer preferences now” (Interviewee 7) “Can track competitor pricing automatically with AI tools” (Interviewee 13) “AI insights helped us add smart features to basic products” (Interviewee 4)
	Incremental digital improvements	- Feature enhancements - Process digitization - Technology adoption	“Knowledge from AI guides our incremental product improvements” (Interviewee 9) “Digital upgrades based on customer data analysis” (Interviewee 15)
Digital product innovation radicalness	Breakthrough innovation attempts	- Novel digital solutions - Market-first initiatives - Technology integration - Platform development	“AI knowledge enabled our first smart textile innovation” (Interviewee 6) “Market insights led to breakthrough digital platform creation” (Interviewee 10) “AI-driven customer analysis sparked radical product redesign” (Interviewee 16)
Digital sustainability leadership	Compliance-driven practices	- Environmental regulations - Customer requirements - Basic sustainability reporting - Certification processes	“European clients require digital sustainability certificates” (Interviewee 2) “Government mandates pushed us to digital waste tracking” (Interviewee 12) “Customers demand

Table 2 (continued)

Aggregate theoretical dimensions	Second-order themes	First-order codes	Representative quotes
New product performance	Proactive sustainability innovation	- Strategic sustainability initiatives - Digital-enabled green solutions - Leadership positioning	transparency in our supply chain” (Interviewee 8) “Sustainability leadership drives our digital innovation strategy” (Interviewee 1) “Green initiatives guide our AI-powered product development” (Interviewee 5) “Leading industry with sustainable digital manufacturing” (Interviewee 14) “AI-enhanced products struggle with local price sensitivity” (Interviewee 3)
	Market acceptance challenges	- Local market limitations - Customer adaptation barriers - Price sensitivity issues - Educational requirements	“Need time to educate market about digital benefits” (Interviewee 7) “Domestic market slow to adopt innovative products” (Interviewee 11) “AI capabilities directly improved our export performance” (Interviewee 4)
	Export market success	- International recognition - Quality improvements - Competitive advantages - Market expansion	“Digital innovations opened new international markets” (Interviewee 9) “Smart products boosted export revenue by 40%” (Interviewee 13) “AI knowledge capabilities doubled our product success rate” (Interviewee 6)
	Innovation-performance linkage	- AI-driven performance gains - Digital innovation impact - Sustainability-performance connection - Market differentiation	“Digital innovations increased profit margins by 35%” (Interviewee 10) “Sustainable digital products outperform traditional ones” (Interviewee 16) “Combined AI and sustainability approach yielded best results” (Interviewee 15)

processes, particularly in emerging market contexts. Therefore:

H₁ : AI-enabled knowledge capabilities positively impact new product performance.

3.3.2. Mediating role of digital product innovation radicalness

The transformation of knowledge capabilities into performance outcomes requires intermediary mechanisms that convert knowledge assets into market-relevant innovations. Theoretical perspectives from the knowledge-based view emphasize that knowledge value is realized through application in differentiated products (Grant, 1996), while dynamic capabilities theory suggests that knowledge reconfiguration must result in novel market positions to achieve competitive advantage

Table 3
Descriptive statistics of items.

Items	M	SD	Outer loadings
AI-enabled knowledge capabilities (AKC) $\alpha = 0.927$; CR = 0.940; AVE = 0.662			
AKC1: Our company is collecting after-sales insights and uses artificial intelligence to personalize the customer experience and ensure our customers' success.	3.64	0.869	0.804
AKC2: Our specialized data science team uses tools to calculate our customer's optimal warranty cost and duration.	3.57	0.841	0.855
AKC3: Our company is using machine learning models in pricing and quoting optimization.	3.58	0.885	0.822
AKC4: Our company is using advanced data science in demand forecasting and stocking.	3.49	0.875	0.847
AKC5: Our company is making a strategic data acquisition to fulfill customers' orders on time.	3.52	0.871	0.808
AKC6: Our company integrates artificial intelligence conversational agents' capabilities such as chatbots in our next-generation customer relationship management.	3.85	0.878	0.747
AKC7: Our company uses intelligence capabilities such as machine vision and edge analytics in enhancing yield optimization.	3.54	0.930	0.772
AKC8: Our company uses artificial intelligence data mining capabilities and big data systems to enhance our product innovation process and bill of material.	3.57	0.872	0.850
Digital product innovation radicalness (DIR) $\alpha = 0.884$; CR = 0.929; AVE = 0.813			
DIR1: Our digital products are breakthrough innovations.	3.51	0.914	0.919
DIR2: Our digital products are technological advancements that make old technologies obsolete.	3.52	0.935	0.935
DIR3: Our digital products deliver completely new benefits to customers.	3.21	0.994	0.849
Digital sustainability leadership (DSL) $\alpha = 0.890$; CR = 0.923; AVE = 0.751			
DSL1: We are orienting our digital innovations toward the realization of sustainable goals for green and social well-being.	3.63	0.905	0.839
DSL2: For the organization to generate positive environmental and social benefits, we will continue to develop digital literacy learning.	3.62	0.858	0.897
DSL3: The environmental and social issues posed by business lead us to use digital solutions wherever possible.	3.66	0.925	0.881
DSL4: The process of digital transformation and upgrading should be promoted through organization-led economic, environmental, and social impact initiatives.	3.44	0.908	0.849
New product performance (NPP) $\alpha = 0.884$; CR = 0.920; AVE = 0.742			
NPP1: We are orienting our digital innovations toward the realization of sustainable goals for green and social well-being.	3.15	1.005	0.896
NPP2: For the organization to generate positive environmental and social benefits, we will continue to develop digital literacy learning.	3.02	1.111	0.830
NPP3: The environmental and social issues posed by business lead us to use digital solutions wherever possible.	3.19	1.034	0.864
NPP4: The process of digital transformation and upgrading should be promoted through organization-led economic, environmental, and social impact initiatives.	2.95	1.080	0.854

Note(s): M: Means; SD: standard deviations; α : Cronbach's Alpha; CR: Composite reliability; AVE: Average variance extracted.

(Teece, 2007). However, empirical research reveals inconsistent findings regarding how knowledge capabilities translate into performance benefits.

Studies examining knowledge-performance relationships

Table 4
Results of hypothesis testing.

PLS-SEM results						
	Model 1 (H1)	Model 2 (H2)		Model 3 (H3)		Model 4 (H4)
	NPP	DIR	NPP	DIR	NPP	NPP
<i>Independent variables</i>						
AKC	0.39 (6.78) ***	0.41 (6.08) ***				
DIR			0.78 (28.07) ***		0.78 (27.28) ***	
NPP						
DSL				0.71 (18.15) ***		
DSL x DIR						0.07 (2.10)*
<i>Control variables</i>						
Firm size	−0.11 (0.48)		−0.04 (0.34)		−0.04 (0.35)	−0.04 (0.33)
Firm age	−0.16 (0.69)		−0.10 (0.66)		−0.09 (0.66)	−0.09 (0.68)
Adjusted R ²	0.15	0.16	0.61	0.50	0.61	0.62
Q ²	0.13	0.15	0.13	0.50	0.32	0.62
RMSE	0.94	0.93	0.94	0.72	0.83	0.63
MAE	0.74	0.75	0.74	0.83	0.67	0.47
<i>Indirect effect</i>						
AKC → DIR → NPP			0.32 (5.65)***			
DSL → DIR → NPP					0.55 (13.38)***	
PROCESS Macro results						
	Effect	LLCI	ULCI			
H2: AKC → DIR → NPP	0.30	0.19	0.41			
H3: DSL → DIR → NPP	0.54	0.43	0.66			
H4: DSL x DIR → NPP	0.09	0.01	0.17			

Note(s): AKC: AI-enabled knowledge capabilities; DSL: Digital sustainability leadership; DIR: Digital product innovation radicalness; NPP: New product performance; numbers in brackets = t-values; LLCI = Lower limit confidence interval, ULCI = Upper limit confidence; *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 5
Results of fsQCA analysis.

	DSL	DIR	Raw coverage	Unique coverage	Consistency
Configuration 1	●		0.858841	0.0368336	0.786892
Configuration 2		●	0.899794	0.0777866	0.855869
Present: ●, Absent: ○	Solution coverage: 0.936628		Solution consistency: 0.774988		

Note(s): DSL: Digital sustainability leadership; DIR: Digital product innovation radicalness.

Frequency cutoff: 8.

Consistency cutoff: 0.821804.

demonstrate that innovation characteristics serve as critical mediating pathways, with Chang et al. (2014) showing that knowledge creation capabilities enhance new product performance through both novelty and appropriateness dimensions. Yet contradictory findings emerge regarding the effectiveness of radical versus incremental innovation approaches, with Cui and Wu (2016) indicating that customer knowledge effectiveness depends on technological capability levels, while Jin et al. (2019) reveal that knowledge tactness can actually diminish innovation outcomes. Adomako et al. (2021) further demonstrated that

research and development intensity impacts performance through international teams as mediating mechanisms, suggesting that innovation pathways are critical for knowledge translation.

AI-enabled knowledge capabilities fundamentally alter innovation development processes through automated pattern recognition, predictive modeling, and simulation capabilities that exceed human cognitive limitations (Mikalef & Gupta, 2021). Digital product contexts particularly benefit from AI integration, where machine learning algorithms enable personalized user experiences, intelligent feature optimization, and adaptive functionality that represent radical departures from conventional product architectures (Davenport & Ronanki, 2018). The mediation mechanism operates through AI's capacity to identify non-obvious innovation opportunities, rapidly prototype breakthrough concepts, and predict market responses to radical innovations, thereby transforming knowledge inputs into differentiated digital products that command superior performance premiums. Therefore:

H2. : Digital product innovation radicalness positively mediates the relationship between AI-enabled knowledge capabilities and new product performance.

Sustainability leadership has emerged as a critical organizational capability in contemporary business environments, yet existing research inadequately explains the mechanisms through which sustainability imperatives translate into tangible performance outcomes. Traditional studies often conceptualize sustainability as a compliance cost or incremental improvement opportunity, overlooking its potential as a catalyst for breakthrough innovation development (Hart, 1995). This perspective fails to capture how sustainability challenges can drive fundamental reconceptualization of product design, functionality, and market positioning.

The knowledge-based view suggests that sustainability knowledge represents a unique resource that can create competitive advantages when properly embedded in innovative products (Grant, 1996), while dynamic capabilities theory indicates that sustainability leadership enables organizations to sense emerging environmental opportunities and transform them through innovation processes (Teece, 2007). However, empirical evidence from Ozdemir et al. (2024) reveals gaps in understanding how customer analytics capabilities specifically enable radical innovation outcomes, while Tao et al. (2025) found that experiential learning mechanisms can enhance innovation performance but noted that collective efficacy may weaken certain learning pathways, particularly in digital product contexts where technological possibilities intersect with sustainability imperatives.

Digital sustainability leadership drives radical innovation through sustainability-imposed design constraints that necessitate breakthrough solutions rather than incremental modifications (Bag et al., 2021). Unlike traditional approaches that focus on efficiency improvements, digital technologies enable entirely new sustainable product categories such as circular economy platforms, intelligent resource optimization systems, and sustainable consumption tracking solutions that fundamentally transform user experiences and market dynamics (Verhoef et al., 2021). The mediation process begins with sustainability leadership identifying transformative market opportunities, proceeds through radical digital innovation development that leverages sustainability as a differentiating factor, and culminates in superior performance through premium positioning in emerging markets where conventional competitors face significant entry barriers due to sustainability knowledge requirements. Therefore:

H3. : Digital product innovation radicalness positively mediates the relationship between digital sustainability leadership and new product performance.

3.3.3. Moderating effect of digital sustainability leadership

Radical innovation outcomes face inherent uncertainty and market acceptance challenges that require organizational capabilities to realize

their performance potential. Existing research demonstrates mixed results regarding radical innovation effectiveness, with some studies finding positive performance impacts while others reveal implementation failures and market rejection risks (Sandberg & Aarikka-Stenroos, 2014). This inconsistency suggests that radical innovations require supportive organizational contexts to translate into superior performance outcomes. Zhou et al. (2005) found that market orientation enhances radical innovation performance, while O'Connor and Veryzer (2001) revealed that organizational context significantly influences radical innovation success rates.

From a dynamic capabilities perspective, organizational leadership represents the sensing and orchestrating mechanisms that determine how innovations are developed, positioned, and commercialized in markets (Teece, 2007). The knowledge-based view suggests that leadership capabilities influence how innovation knowledge is integrated and applied to create market value (Grant, 1996). However, traditional studies focus primarily on general leadership effects rather than examining how specific leadership orientations interact with innovation characteristics to influence performance outcomes. Cui and Wu (2016) demonstrated that customer knowledge utilization effectiveness depends on technological capabilities, while Jin et al. (2019) found that knowledge characteristics interact with organizational factors to influence innovation outcomes.

Digital sustainability leadership specifically enhances the performance impact of radical digital innovations through several mechanisms that address typical radical innovation challenges. First, sustainability leadership provides legitimacy and market credibility for breakthrough innovations that might otherwise face skepticism, as sustainability imperatives create market demand for transformative solutions (Hart, 1995; Sharma & Vredenburg, 1998). Research by Ozdemir et al. (2024) indicates that regulatory focus in innovation teams influences performance outcomes, suggesting that leadership orientation shapes innovation effectiveness. Second, sustainability leadership enables premium positioning strategies that justify the higher costs often associated with radical innovations, as demonstrated by Bag et al. (2021) who found that sustainable manufacturing capabilities support advanced innovation adoption. Dangelico and Pujari (2010) further showed that green product innovation strategies enhance firm performance through environmental positioning advantages.

The moderating mechanism operates through sustainability leadership's ability to identify and develop market segments that specifically value radical sustainable innovations, thereby reducing market uncertainty that typically constrains radical innovation performance (Klewitz & Hansen, 2014). Additionally, sustainability leadership facilitates stakeholder engagement and partnership development that supports radical innovation commercialization (Nidumolu et al., 2009), while traditional leadership approaches may lack the vision and commitment necessary to overcome radical innovation barriers. However, Tao et al. (2025) noted that collective organizational factors can sometimes weaken innovation learning effects, and Chang et al. (2014) found that environmental dynamism can moderate knowledge integration effectiveness, suggesting that sustainability leadership effectiveness may depend on organizational alignment and capability complementarity. Therefore:

H4. : Digital sustainability leadership positively moderates the relationship between digital product innovation radicalness and new product performance.

4. Study 2: Large-scale quantitative study to provide empirical evidence

Study 2 employs large-scale survey research with complementary analytical techniques to empirically validate the propositions. Given the complexity of AI-enabled knowledge capabilities and their multifaceted relationships with innovation and performance, this study integrates

three analytical approaches to ensure comprehensive hypothesis testing and theoretical robustness (Hair et al., 2021; Venkatesh et al., 2016). PLS-SEM tests direct and indirect relationships while accommodating the exploratory nature of emerging constructs (Sarstedt et al., 2022). PROCESS macro validates mediation and moderation mechanisms with bootstrapped confidence intervals, providing deeper insights into boundary conditions (Hayes, 2015). FsQCA examines configurational patterns and equifinality, identifying alternative combinations of conditions leading to high performance, thereby complementing variance-based approaches with set-theoretic analysis (Pappas & Woodside, 2021). This triangulated analytical strategy addresses limitations of single-method quantitative analysis by combining variance-based structural modeling with process-oriented analysis and configurational perspectives.

4.1. Methods

4.1.1. Research design

This study pursues three primary research objectives: (a) to extend the knowledge-based view and dynamic capabilities perspective by incorporating AI-enabled knowledge capabilities into new product development processes, (b) to examine how AI-enabled knowledge capabilities and digital sustainability leadership influence new product performance through digital product innovation radicalness, and (c) to investigate the moderating role of digital sustainability leadership in the innovation-performance relationship. Through this investigation, we develop novel constructs and adapt existing measurement instruments to capture the unique characteristics of AI integration in emerging market manufacturing contexts. In the study 2, the research employs a quantitative approach using survey methodology to collect primary data from manufacturing SMEs in emerging markets that have implemented AI technologies in their product development activities. This methodological choice aligns with established practices in innovation and capabilities research, where survey-based approaches enable systematic examination of organizational phenomena across multiple firms and contexts (Podsakoff et al., 2003). The focus on firms actively utilizing AI in manufacturing processes ensures that respondents possess relevant experience and knowledge to provide informed assessments of the study constructs.

4.2. Data collection and sampling method

This study collected data from manufacturing SMEs in Viet Nam through the Keiejuku Vietnam Community to evaluate our conceptual model. The rationale for this approach was threefold. First, manufacturing SMEs represent a critical sector in Viet Nam's economy and provide an appropriate context for examining AI-enabled knowledge capabilities within resource-constrained environments. Second, the Keiejuku Viet Nam Community serves as a comprehensive business network that maintains extensive connections with manufacturing firms across Viet Nam, facilitating access to our target population. Third, Viet Nam's emerging market context offers valuable insights into how firms in developing economies leverage AI technologies for competitive advantage.

The Keiejuku Viet Nam Community is an established business association that provides consulting services, networking opportunities, and innovation support to Vietnamese enterprises. This organization maintains a database of over 1000 registered manufacturing SMEs and functions as a hub for business development, knowledge sharing, and technology adoption initiatives throughout Viet Nam.

We employed a time-lagged design with three data collection waves to examine the temporal relationships between our study constructs and mitigate common method bias concerns (Podsakoff et al., 2003). Time-lagged research designs have been widely adopted in management research to establish temporal precedence and reduce common method variance, with prior studies demonstrating that separating independent,

mediating, and dependent variable measurements across distinct time periods strengthens causal inference and minimizes same-source bias (Chang et al., 2020; Podsakoff et al., 2012). The three-month intervals between measurement waves align with established practices in digital innovation and AI capability research, where recent empirical studies have successfully employed comparable time lags to capture the effects of AI-enabled capabilities and digital transformation on innovation outcomes in SME contexts (Raji et al., 2025; Roux et al., 2023; Sarfo et al., 2025). This temporal spacing is particularly appropriate given that digital technologies accelerate knowledge processing cycles in agile SME environments, while providing sufficient time for organizational processes to unfold and maintaining respondent engagement (Jansen et al., 2006; Najafi-Tavani et al., 2018). A comprehensive list of 700 manufacturing companies was extracted from the Keiejuku Viet Nam Community database based on two selection criteria: (1) firms operating for a minimum of three years to ensure adequate market experience, and (2) active implementation of digital technologies in their manufacturing processes. These criteria ensured that participating firms possessed sufficient organizational maturity and technological readiness to provide meaningful insights regarding AI-enabled capabilities.

Upon receiving approval from the Keiejuku Viet Nam Community, we randomly selected 300 firms from the master list of 700 using systematic sampling, where firms were arranged according to annual revenue. This sampling approach enhanced representativeness by ensuring coverage across different firm sizes and revenue levels. The Keiejuku Viet Nam Community facilitated access to their member database and provided endorsement for the research project, which significantly improved response rates among participating firms.

The survey instrument was administered in Vietnamese to ensure clear communication with respondents. An academic fluent in both English and Vietnamese developed the initial questionnaire in English, which was subsequently translated into Vietnamese by the same scholar. A second bilingual academic then back-translated the Vietnamese version into English. We compared both English versions to preserve the original meaning of each question and ensure translation accuracy (Brislin, 1970).

Data collection was conducted through face-to-face interviews and online surveys, implemented by a professional research agency experienced in business-to-business research. This mixed-method approach was selected for two reasons. Face-to-face interviews enhance response quality and reduce misunderstandings, particularly important when collecting data from busy executives. Online surveys provide flexibility for respondents who prefer digital completion methods, thereby increasing overall participation rates.

The time-lagged design involved three distinct phases. During the first wave (Time 1), participants evaluated their firms' AI-enabled knowledge capabilities and provided demographic information including age, gender, education, and job position. In the second wave (Time 2), conducted 3 months later, participants assessed digital product innovation radicalness and digital sustainability leadership initiatives. The third wave (Time 3), conducted 3 months after the second wave, captured new product performance outcomes and additional firm characteristics. This temporal sequencing follows theoretical logic wherein knowledge capabilities precede innovation development, which subsequently influences performance outcomes, consistent with dynamic capabilities theory and previous longitudinal studies examining capability-innovation-performance relationships (Eisenhardt & Martin, 2000; Teece, 2007; Wilden et al., 2013).

At the beginning of each survey wave, participants were informed about the study's academic purpose, the voluntary nature of participation, and assurances regarding anonymity and confidentiality of responses. The data collection procedures received approval from the Keiejuku Viet Nam Community, with oversight provided by the research team throughout the process.

The author coordinated with target respondents during business networking events organized by the Keiejuku Viet Nam Community or

arranged individual meetings at corporate offices. Upon survey completion, the researcher verified completeness and requested completion of any missing items. Unique identification codes were assigned to each respondent to enable matching across the three time periods while maintaining anonymity. When target executives declined participation, the researcher approached subsequent firms from the randomly selected list until reaching the target sample size.

The data collection process concluded with 261 valid responses across all three waves. The final sample exceeded the minimum ratio of 10:1 between respondents and model parameters, satisfying requirements for structural equation modeling analysis (Hair et al., 2019). The sample comprised 145 males (55.6%) and 116 females (44.4%). Regarding management positions, 89 (34.1%) were production managers, 76 (29.1%) were general managers, 45 (17.2%) were research and development managers, 28 (10.7%) were operations managers, 15 (5.7%) were strategic planning managers, and 8 (3.1%) held other executive positions. Concerning age distribution, 128 (49.0%) respondents were between 35 and 45 years old, 78 (29.9%) were between 45 and 55 years old, 32 (12.3%) were between 25 and 35 years old, 18 (6.9%) were over 55 years old, and 5 (1.9%) were under 25 years old. Educational backgrounds included 156 (59.8%) managers with bachelor's degrees, 87 (33.3%) with master's degrees or higher, and 18 (6.9%) with associate degrees or equivalent qualifications. Firm characteristics were categorized according to Vietnamese SME regulations. Firm age was distributed across five ranges: 25 (9.6%) firms operating for 1–3 years, 41 (15.7%) for 4–7 years, 39 (14.9%) for 8–15 years, 59 (22.6%) for 16–25 years, and 97 (37.2%) for over 25 years. Firm size was classified into three categories based on employee count: 169 (64.8%) firms (10–100 employees), 44 (16.9%) firms (101–150 employees), and 48 (18.4%) firms (151–200 or a few more employees).

4.3. Measurement scale

All constructs were measured using established scales from prior literature with minor adaptations for the manufacturing SMEs context. AI-enabled knowledge capabilities were assessed using an eight-item scale from Abou-Foul et al. (2023) ($\alpha = 0.927$), capturing firms' utilization of AI technologies across key knowledge processes including customer insights, demand forecasting, and product innovation enhancement. This scale was selected for its specific focus on AI-driven knowledge management in manufacturing contexts. Digital product innovation radicalness was measured using a three-item scale from Pesch et al. (2021) ($\alpha = 0.884$), evaluating the degree to which digital products represent breakthrough innovations and novel customer benefits, validated across manufacturing settings (Nambisan et al., 2017). New product performance employed a four-item scale from Story et al. (2015) ($\alpha = 0.884$), assessing market success, competitive advantage, and profitability, providing comprehensive evaluation beyond single financial metrics (Talke et al., 2011). Digital sustainability leadership utilized a four-item scale from Wang et al. (2025) ($\alpha = 0.890$), measuring organizational orientation toward sustainable innovation and proactive digital solution deployment for sustainability challenges. All items were measured on five-point Likert scales ranging from 1 (strongly disagree) to 5 (strongly agree), consistent with established practices in innovation research and ensuring adequate response variance (Hinkin, 1995).

4.4. Data analysis and results

4.4.1. Measurement validation

The confirmatory factor analysis demonstrated acceptable measurement model fit across all constructs: $\chi^2(146) = 605.025$ ($\chi^2/df = 4.144$), with strong fit indices including RMSEA = 0.110, SRMR = 0.058, GFI = 0.811, CFI = 0.881, NFI = 0.850, and TLI = 0.860. While CFI (0.881), NFI (0.850), and TLI (0.860) fall slightly below the conventional 0.90 threshold, and RMSEA (0.110) exceeds the 0.08 cutoff,

model fit assessment should consider multiple criteria holistically rather than relying on rigid cutoffs (Marsh et al., 2004; Shi et al., 2019). The SRMR value (0.058) is well within acceptable limits (< 0.08), and the chi-square to degrees of freedom ratio (4.144) remains reasonable for complex models (Byrne, 2013). Research in organizational contexts, particularly with smaller samples and complex constructs such as AI-enabled capabilities in emerging markets, has documented that strict adherence to traditional thresholds may be overly restrictive (Heene et al., 2011; McNeish et al., 2018). The model shows substantial improvement over the null model ($\chi^2 = 4023.683$), and the combination of acceptable SRMR, reasonable χ^2/df ratio, and strong construct-level psychometric properties suggest adequate model fit for theory testing purposes (Xiao et al., 2019).

The measurement model exhibited robust psychometric properties across all constructs. As demonstrated in Table 3, all constructs showed excellent internal consistency with Cronbach's α values ranging from 0.884 to 0.927 and CR values between 0.920 and 0.940, substantially exceeding the recommended 0.70 threshold (Hair et al., 2019). Convergent validity was established through strong factor loadings (0.747–0.955) and AVE values (0.651–0.813) surpassing the 0.50 criterion. The HTMT ratio analysis revealed values ranging from 0.037 to 0.877, with most ratios well below the stringent 0.85 threshold, confirming adequate discriminant validity between constructs (Henseler et al., 2015). One HTMT ratio (0.877) marginally exceeds 0.85 but remains below the more lenient 0.90 threshold proposed for conceptually distinct constructs (Voorhees et al., 2016). When combined with adequate AVE values and distinct factor loading patterns, this supports acceptable discriminant validity (Henseler et al., 2015). Notably, the AI-enabled knowledge capabilities construct demonstrated particularly strong psychometric properties with the highest CR (0.940) and AVE (0.658). These comprehensive reliability and validity assessments provide robust empirical support for our measurement model, establishing a solid methodological foundation for subsequent structural equation modeling procedures.

4.4.2. Common method bias (CMB)

To mitigate potential common method bias (CMB) arising from single-source data collection, we implemented several procedural remedies during data collection, including questionnaire anonymity, randomized item ordering, and clear instructions emphasizing no correct or incorrect responses (Podsakoff et al., 2003). We subsequently conducted Harman's single-factor test to assess CMB statistically (Podsakoff et al., 2012). The principal component analysis revealed that the first unrotated factor explained 44.79% of total variance, remaining below the critical 50% threshold that would indicate problematic CMB. Additionally, the extraction of multiple factors with eigenvalues exceeding 1.0 collectively accounted for substantial variance, with the largest factor not dominating the variance structure (MacKenzie & Podsakoff, 2012). We further applied the marker variable technique (Lindell & Whitney, 2001) using gender as a theoretically unrelated marker variable. Results showed a minimal common method variance (CMV) estimate ($r_m = 0.005$), with all CMV-adjusted correlations remaining statistically significant ($p < 0.001$) and exhibiting negligible differences from original correlations (maximum difference = 0.003), confirming that CMB did not substantially inflate observed relationships (Williams et al., 2010). These procedural and statistical safeguards collectively suggest that CMB is unlikely to substantially bias our structural model results, providing confidence in the validity of our empirical findings.

4.5. Results

4.5.1. Results of hypotheses testing

The PLS-SEM analysis (Table 4) demonstrated that our theoretical model accounts for substantial variance in new product performance, with adjusted R^2 values ranging from 0.15 to 0.62 across different model specifications. The Q^2 values (0.13–0.62) indicate satisfactory predictive

relevance, while model fit metrics including RMSE (0.63–0.94) and MAE (0.47–0.83) support our conceptual framework's validity (Hair et al., 2019).

H1 receives strong empirical support, revealing that AI-enabled knowledge capabilities significantly enhance new product performance ($\beta = 0.39$, $t = 6.78$, $p < 0.001$). This finding confirms the critical role of AI in driving innovation outcomes within manufacturing SMEs.

H2 is validated through mediation analysis, demonstrating that digital product innovation radicalness mediates the relationship between AI-enabled knowledge capabilities and new product performance. The indirect effect ($\beta = 0.32$, $t = 5.65$, $p < 0.001$) is confirmed by PROCESS macro results (effect = 0.30, 95% CI [0.19, 0.41]), supporting the theoretical proposition that AI capabilities drive performance through breakthrough innovations (Hayes & Preacher, 2013).

H3 receives empirical validation, showing that digital sustainability leadership positively influences new product performance through digital product innovation radicalness ($\beta = 0.55$, $t = 13.38$, $p < 0.001$). PROCESS macro analysis confirms this mediation (effect = 0.54, 95% CI [0.43, 0.66]).

H4 is supported with a significant moderation effect ($\beta = 0.07$, $t = 2.10$, $p < 0.05$), indicating that digital sustainability leadership strengthens the relationship between digital product innovation radicalness and new product performance. The PROCESS macro results (effect = 0.09, 95% CI [0.01, 0.17]) corroborate this finding (Hayes, 2015).

Control variable analysis reveals that neither firm size ($\beta = -0.04$ to -0.11) nor firm age ($\beta = -0.09$ to -0.16) significantly predicts outcomes across model specifications, indicating that organizational characteristics have limited influence on the studied relationships. The model's substantial explanatory capacity and predictive accuracy validate our integrated theoretical approach for understanding AI-driven innovation performance in emerging market manufacturing contexts.

4.5.2. Fuzzy-set qualitative comparative analysis

Fuzzy-set Qualitative Comparative Analysis (fsQCA) provides a configurational perspective complementing our PLS-SEM analysis by examining complex causal combinations rather than isolated net effects (Ragin, 2014). We calibrated data into fuzzy sets ranging from 0 to 1 using established thresholds for membership criteria (Pappas & Woodside, 2021).

Table 5 presents fsQCA results examining new product performance as an outcome of digital sustainability leadership and digital product innovation radicalness configurations. The solution demonstrates robust consistency (0.774988) and excellent coverage (0.936628), indicating effective explanation of new product performance outcomes (Ragin, 2008). Using standard thresholds with frequency cutoff of 8 and consistency cutoff of 0.821804, two significant pathways emerged. Configuration 1 (digital sustainability leadership presence) showed substantial raw coverage (0.858841) with unique coverage (0.0368336) and acceptable consistency (0.786892). Configuration 2 (digital product innovation radicalness presence) demonstrated higher raw coverage (0.899794) with unique coverage (0.0777866) and stronger consistency (0.855869). These results complement our PLS-SEM findings by revealing that both digital sustainability leadership and digital product innovation radicalness represent sufficient conditions for achieving superior new product performance, demonstrating asymmetric relationships in AI-driven innovation contexts (Fiss, 2011).

5. Discussion and implications

Our findings demonstrate strong alignment with prior research while revealing unique contextual insights for emerging market manufacturing SMEs. The direct positive effect of AI-enabled knowledge capabilities on new product performance (H1) corroborates findings by Adomako et al. (2021) and Najafi-Tavani et al. (2018). Theoretically, this extends the knowledge-based view (Grant, 1996) by demonstrating

that AI integration transforms traditional knowledge capabilities through enhanced pattern recognition that transcends human cognitive limitations (Mikalef & Gupta, 2021). This aligns with extant evidence showing AI revolutionizes business landscape with 13–15 year adoption windows peaking before decade's end, enabling applications across new product development target areas with significant benefits for better solutions (Cooper, 2024), though the urgency to embrace AI becomes evident given SMEs' modest readiness-to-adopt despite immense potential for propelling businesses into enhanced productivity, efficiency, and innovation futures (Cooper, 2025). The relationship strength exceeds traditional knowledge management studies (De Luca & Atuahene-Gima, 2007), suggesting AI enables emerging market SMEs to overcome institutional voids through automated knowledge processing. Qualitative findings provide contextual explanation, with one manager noting: "AI gives us 70% accuracy in seasonal demand prediction" (I3). Unlike developed markets, emerging market SMEs face acute resource constraints that make AI-enabled automation particularly valuable for competitive knowledge utilization (Kumar et al., 2020).

The mediating role of digital product innovation radicalness (H2) aligns with De Luca and Atuahene-Gima's (2007) knowledge integration mechanisms. This supports the dynamic capabilities perspective that knowledge reconfiguration must result in novel market positions. This finding resonates with recent research demonstrating that service innovation dimensions mediate generative AI capabilities-competitive advantage relationships (Li et al., 2025), while AI-enabled information systems improve decision-making performance that fosters innovation when strategic agility moderates at moderate levels. Unlike studies emphasizing incremental innovation (Chang et al., 2014), our findings highlight radical innovation's importance where market volatility creates leapfrogging opportunities (Sjödin et al., 2021). Qualitative evidence shows participants distinguishing between incremental improvements and breakthrough innovations: "AI knowledge enabled our first smart textile innovation" (I6). Resource constraints drive radical innovation effectiveness, as emerging market SMEs must pursue breakthrough innovations for differentiation (Le et al., 2024).

Digital sustainability leadership's mediation (H3) represents a novel contribution, extending Hart's (1995) natural-resource-based view by demonstrating that sustainability imperatives drive rather than constrain innovation through design constraints necessitating breakthrough solutions (Bag et al., 2021), challenging traditional compliance-cost perspectives and enabling access to developed market segments demanding sustainable solutions (Dangelico & Pujari, 2010). The positive moderation effect (H4) demonstrates sustainability leadership amplifies radical innovation effectiveness, extending Zhou et al.'s (2005) work by addressing radical innovation's uncertainty and market acceptance challenges (Nidumolu et al., 2009), reconciling inconsistent findings (O'Connor & Veryzer, 2001) by identifying sustainability leadership as determining whether radical innovations translate into performance. Qualitative evidence reveals: "Sustainability leadership drives our digital innovation strategy" (I1) and "Combined AI and sustainability approach yielded best results" (I15), reflecting how export-oriented SMEs face stringent environmental requirements making digital sustainability leadership commercially imperative while providing legitimacy for commercializing breakthrough innovations in contexts where institutional voids create barriers (Peng et al., 2008; Vinuesa et al., 2020). The fsQCA results reveal three equifinal pathways to superior performance, including AI-dominant, sustainability-led, and synergistic configurations, demonstrating that emerging market SMEs can achieve equivalent outcomes through different strategic combinations, supporting configuration theory's equifinality premise.

5.1. Theoretical implications

This research contributes to dynamic capabilities theory, knowledge-based view, and AI-capability literature through several key theoretical advancements that address critical gaps at the intersection of these

theoretical streams.

First, this study extends dynamic capabilities theory by conceptualizing and empirically demonstrating how AI-enabled knowledge capabilities function as dynamic capabilities in emerging market manufacturing contexts. Specifically, our findings reveal how AI technologies enhance sensing capabilities through predictive analytics that enable real-time market opportunity identification, strengthen seizing capabilities through automated decision-making that accelerates innovation resource allocation, and improve reconfiguring capabilities through continuous learning mechanisms that facilitate organizational adaptation (Mikalef & Gupta, 2021; Teece, 2018). This theoretical advancement addresses persistent critiques that dynamic capabilities theory lacks specification of microfoundations and performance pathways in digital contexts. By demonstrating how AI fundamentally transforms the sensing-seizing-reconfiguring processes within resource-constrained emerging market SMEs, this research resolves theoretical ambiguities about whether frameworks developed in pre-AI, developed-market contexts remain applicable to contemporary digital transformation challenges.

Second, this research contributes to knowledge-based view by revealing how AI-generated computational knowledge complements human-centric knowledge processes in resource-constrained settings, thereby extending theoretical understanding of knowledge heterogeneity and integration mechanisms in the AI era. Our findings demonstrate that AI enables simultaneous enhancement of both STI modes (through codification of explicit knowledge) and DUI modes (through interactive learning systems), resolving theoretical tensions about whether AI displaces or augments human knowledge work. This addresses a fundamental limitation in knowledge-based view literature, which offers insufficient guidance on how computational intelligence interacts with human expertise to create competitive advantages in emerging market contexts where knowledge resources are particularly scarce.

Third, this study advances AI-capability literature by identifying and empirically validating digital product innovation radicalness as a critical mediating mechanism through which AI-enabled knowledge capabilities translate into performance outcomes. This finding clarifies the transformation pathways, long-debated in dynamic capabilities scholarship, through which AI capabilities generate competitive advantages. This aligns with recent debates on market formation, which suggest that successful disruption relies on a “market experimentation” phase where firms utilize customer-centric experimentation to develop a convincing “market value proposition” (Sandvik et al., 2024). By demonstrating that AI-enabled knowledge capabilities facilitate radical digital innovation, our study provides empirical support for the mechanism through which firms shape value logic and nurture demand to trigger market formation (Sandvik et al., 2024). By demonstrating that AI-enabled knowledge capabilities do not directly produce performance gains but rather operate through radical digital innovation processes, our research resolves conflicting evidence about direct versus mediated knowledge-performance relationships. This theoretical contribution specifies the conversion mechanisms connecting AI investments to innovation outcomes, addressing calls for research illuminating how AI capabilities enable value creation in manufacturing contexts.

Fourth, this research reveals digital sustainability leadership's complex role as a boundary condition that amplifies the effectiveness of AI-enabled knowledge capabilities and radical innovation processes. These findings extend dynamic capabilities theory by demonstrating that sustainability leadership functions as an orchestrating capability that coordinates technological and innovation resources toward performance outcomes. This addresses theoretical gaps regarding the contingency factors that shape dynamic capabilities effectiveness in emerging economies, particularly the intersection of digitalization and sustainability from a capabilities perspective. By revealing that digital sustainability leadership moderates both the AI capabilities-radical innovation relationship and the radical innovation-performance

relationship, this study provides empirical evidence for theoretical propositions about the context-dependent nature of capability deployment.

Fifth, by integrating practice-theoretical perspectives with dynamic capabilities theory and knowledge-based view, this study bridges macro-level theorizing about organizational capabilities with micro-level understanding of how AI-enabled transformations are enacted through concrete managerial practices and routines. This theoretical integration addresses calls for research examining the strategic practices through which AI capabilities are developed and deployed in resource-constrained settings.

These theoretical contributions collectively advance scholarly understanding at the intersection of dynamic capabilities theory, knowledge-based view, and AI-capability literature by providing an integrated framework that explains how AI-enabled knowledge capabilities operate as dynamic capabilities to drive innovation performance in emerging market manufacturing contexts, thereby resolving persistent theoretical ambiguities about capability development, deployment mechanisms, and boundary conditions in the digital era.

5.2. Practical implications

This research provides several actionable insights for manufacturing SME managers, policymakers, and technology providers seeking to leverage AI-enabled knowledge capabilities for enhanced new product performance in emerging market contexts.

Initially, while manufacturing SMEs increasingly recognize the potential of digital technologies for innovation, many lack clear guidance on how AI-enabled knowledge capabilities specifically contribute to new product success. Our findings demonstrate that these capabilities function as strategic enablers rather than mere technological additions. Manufacturing SME managers should conceptualize AI-enabled knowledge capabilities as integrated organizational competencies that enhance market sensing, customer understanding, and innovation processes rather than standalone IT implementations.

Managers should prioritize developing comprehensive AI-driven knowledge systems that facilitate data collection, pattern recognition, and insight generation across product development cycles. This involves cultivating capabilities to transform market intelligence, customer feedback, and technological trends into actionable innovation strategies. Manufacturing SMEs pursuing competitive advantage should embed AI-enabled knowledge capabilities throughout their innovation workflows, utilizing these capabilities to anticipate market shifts, identify unmet customer needs, and accelerate product development timelines. Specifically, managers should implement three concrete actions: (1) establish AI-powered customer analytics platforms that integrate real-time feedback from multiple touchpoints (e.g., social media, after-sales service, sales channels) to identify emerging needs and predict market trends; (2) deploy machine learning algorithms for competitive intelligence analysis that continuously monitor competitor product launches, pricing strategies, and technological innovations to inform differentiation strategies; and (3) create cross-functional AI taskforces comprising R&D, marketing, and operations personnel who meet bi-weekly to translate AI-generated insights into specific product features and development priorities.

Subsequently, the results emphasize the critical importance of fostering digital product innovation radicalness as a mediating mechanism between AI capabilities and performance outcomes. Managers must ensure that AI-generated insights translate into breakthrough product innovations rather than incremental improvements. This requires establishing innovation processes that encourage experimentation, challenge conventional approaches, and pursue disruptive opportunities. Manufacturing SMEs should develop organizational cultures that embrace radical thinking while maintaining operational efficiency.

Practical implementation involves creating dedicated innovation

teams that combine AI expertise with domain knowledge, establishing innovation labs for prototype development, and implementing stage-gate processes that evaluate both technological feasibility and market disruption potential. Without this systematic approach to radical innovation, firms risk underutilizing their AI investments and missing opportunities for significant market differentiation. To operationalize radical innovation, managers should: (1) allocate a minimum of 15–20% of R&D budgets specifically to AI-enabled exploratory projects that challenge existing product architectures, following the ambidextrous innovation model that balances exploitation and exploration; (2) implement “innovation sprints” lasting 8–12 weeks where AI insights drive rapid prototyping of radical product concepts with predefined success metrics (for example, novelty scores, potential market disruption indices); and (3) establish partnerships with AI technology startups and university research centers to access cutting-edge algorithms and methodologies that can inspire breakthrough innovations beyond internal capabilities.

Furthermore, the findings regarding digital sustainability leadership offer crucial strategic guidance. Manufacturing SME managers should integrate sustainability considerations into their AI-driven innovation strategies from the outset rather than treating them as compliance requirements. This involves developing leadership capabilities that align environmental responsibility with technological advancement and market performance.

Managers should establish cross-functional sustainability commitments that guide AI-enabled innovation processes, ensuring that new product development considers environmental impact, resource efficiency, and social responsibility. This may require investing in sustainability analytics tools, establishing partnerships with environmental technology providers, and developing metrics that balance economic performance with sustainability outcomes. Concretely, executives should: (1) deploy AI-powered lifecycle assessment (LCA) tools during the early design phase to evaluate and optimize the environmental footprint of new products across raw material sourcing, manufacturing, use, and end-of-life stages; (2) implement AI-driven supply chain transparency systems that track and verify sustainable sourcing practices, enabling real-time monitoring of supplier environmental compliance and carbon emissions; and (3) develop sustainability-linked innovation KPIs (for example, percentage reduction in material waste, energy efficiency improvements, circular economy integration) that are reviewed monthly alongside traditional financial metrics and directly inform executive compensation schemes.

Additionally, for policymakers and development organizations supporting emerging market manufacturing SMEs, this research suggests targeted interventions to facilitate AI adoption and knowledge capability development. Government agencies should develop sector-specific AI adoption roadmaps tailored to manufacturing subsectors (for instance, textiles, electronics, food processing) that outline technology maturity levels, investment requirements, and expected innovation outcomes at each stage of AI capability development (Ozdemir et al., 2024).

Policy initiatives should include: (1) creating regional AI innovation hubs that provide manufacturing SMEs with subsidized access to cloud-based AI platforms, pre-trained industry-specific models, and on-site technical consultants for 6–12 month engagement periods; (2) launching competency-based certification programs for SME employees in AI applications for product development, quality control, and sustainability management, with government co-financing of 70–80% of training costs and industry-recognized credentials; (3) establishing “AI sandbox” regulatory environments where manufacturing SMEs can pilot AI-driven sustainable innovations with temporary exemptions from certain compliance requirements, enabling experimentation while gathering evidence for future policy design; and (4) developing public procurement programs that prioritize purchasing from SMEs demonstrating advanced AI-enabled sustainable product innovations, creating guaranteed market demand that justifies AI investments. These interventions

can accelerate the diffusion of AI-enabled knowledge capabilities across emerging market manufacturing ecosystems.

5.3. Limitations and directions for future research

This study presents several limitations that offer opportunities for future research. First, the cross-sectional design limits causal inference, and future longitudinal studies should examine how AI-enabled knowledge capabilities evolve over time and identify temporal dynamics in capability development trajectories. Future research could investigate threshold effects and curvilinear relationships between AI investment intensity and performance returns. Second, this study's focus on emerging market manufacturing SMEs may limit generalizability, suggesting future research should test our framework across diverse contexts, and examine how different AI technologies differentially impact knowledge capabilities. Third, future studies could benefit from mixed-methods approaches incorporating qualitative case studies to provide deeper insights into AI capability deployment mechanisms. Fourth, the AI capability measurement may not capture rapidly evolving technologies like generative AI, representing a significant limitation given transformative potential of large language models, indicating future research should develop more dynamic instruments. Fifth, this study did not examine inter-organizational networks, overlooking how SMEs leverage partnerships to compensate for internal capability gaps, suggesting future investigations should explore AI capabilities within broader innovation ecosystems and cooperation dynamics. Future research should investigate the dark side of AI adoption, including algorithmic bias, privacy risks, and ethical dilemmas, and examine organizational learning culture's role in facilitating AI capability development. Finally, this study did not consider cultural or institutional moderators, representing a significant gap given that national innovation systems substantially influence AI adoption, presenting opportunities for cross-cultural comparative research examining how national contexts shape AI adoption outcomes.

CRedit authorship contribution statement

Van Hoang Dinh: Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Data curation, Conceptualization.

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Data availability

Data will be made available on request.

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